

## Software Requirements Specifications of SCMS software

Doc # SCMS-SRS

Version: 1.5

Page 1 / 12

### REVISION HISTORY

| Date       | Version | Description                                                                                             | Author                    |
|------------|---------|---------------------------------------------------------------------------------------------------------|---------------------------|
| 14/03/2026 | 1.0     | Initial SRS draft created; completed Introduction and system overview                                   | Emir                      |
| 14/03/2026 | 1.1     | Added User Requirements (Section 2.1) and drafted text for Use Cases.                                   | Bora                      |
| 14/03/2026 | 1.2     | Created and integrated System Architecture and Usage Scenario diagrams. Completed sections 2.2 and 2.3. | Hamza, Emir               |
| 14/03/2026 | 1.3     | Formatted System Requirements (Section 2.4) and aligned traceability IDs.                               | Genidy                    |
| 16/03/2026 | 1.4     | Final team review                                                                                       | Emir, Bora, Hamza, Genidy |
| 23/03/2026 | 1.5     | Corrected diagrams based on feedbacks                                                                   | Hamza, Genidy, Emir, Bora |

| Software Requirements Specifications of SCMS software |              |             |
|-------------------------------------------------------|--------------|-------------|
| Doc # SCMS-SRS                                        | Version: 1.5 | Page 2 / 12 |

## **TABLE OF CONTENTS**

|                                       |          |
|---------------------------------------|----------|
| <b>1 INTRODUCTION</b>                 | <b>3</b> |
| 1.1 Document overview                 | 3        |
| 1.2 System overview                   | 3        |
| 1.3 Abbreviations                     | 3        |
| 1.4 References                        | 3        |
| 1.5 Conventions                       | 3        |
| <b>2 REQUIREMENTS</b>                 | <b>5</b> |
| 2.1 User Requirements Specification   | 5        |
| 2.2 System Architecture               | 5        |
| 2.3 Use Cases and Usage Scenarios     | 6        |
| 2.4 System Requirements Specification | 11       |

| Software Requirements Specifications of SCMS software |              |             |
|-------------------------------------------------------|--------------|-------------|
| Doc # SCMS-SRS                                        | Version: 1.5 | Page 3 / 12 |

## 1 INTRODUCTION

### 1.1 Document overview

This document presents the software requirements specifications of **SCMS** software development project. It describes functional requirements, quality requirements, constraints, use cases, and usage scenarios. It also provides an overview of the system architecture.

### 1.2 System overview

The Student Club Management System (SCMS) is a centralized, Java-based desktop application designed to streamline the administration of university student organizations. It is intended to automate administrative work for university student clubs. At its core, the application will deal with member registration, event organization, and financial management. It is expected to provide a common platform for both the presidents of the clubs (Administrators) and students (Members) to communicate effectively. By automating these interactions, the SCMS reduces manual paperwork and ensures that club operations are transparent and well-organized.

### 1.3 Abbreviations

SCMS: Student Club Management System  
UML: Unified Modeling Language  
IDE: Integrated Development Environment  
JDK: Java Development Kit  
GUI: Graphical User Interface  
SQL: Structured Query Language  
DAO: Data Access Object  
SRS: Software Requirement Specification  
STP: Software Test Plan  
SDD: Software Design Document  
STR: Software Test Report  
SAR: Software Analysis Report

### 1.4 References

| #    | Document Identifier | Document Title                    |
|------|---------------------|-----------------------------------|
| [R1] | SCMS-SDP-v1.5       | Software Development Plan of SCMS |
| [R2] | SCMS-STP            | Software Test Plan of SCMS        |
| [R3] | SCMS-SDD            | Software Detailed Design of SCMS  |

### 1.5 Conventions

Requirements listed in this document are constructed according to the following structure:

---

**User requirement:**

#### **SRS-SCMS-001**

Description of high-level user needs.

**System requirement (traceable to user requirement):**

| Software Requirements Specifications of SCMS software |              |             |
|-------------------------------------------------------|--------------|-------------|
| Doc # SCMS-SRS                                        | Version: 1.5 | Page 4 / 12 |

### **SRS-SCMS-001.1**

Description of the SRS-SCMS-001.1 system requirement, which is related to the user requirement SRS-SCMS-001.

Every system requirement ID includes the ID of the corresponding user requirement as a prefix.

| Software Requirements Specifications of SCMS software |              |             |
|-------------------------------------------------------|--------------|-------------|
| Doc # SCMS-SRS                                        | Version: 1.5 | Page 5 / 12 |

## 2 REQUIREMENTS

### 2.1 User Requirements Specification

The following requirements represent the high-level functional needs of the system's primary users: Club Administrators (Presidents) and General Members.

**SRS-SCMS-001: Member Registration** The SCMS shall enable Club Administrators to create, read, update, and delete (CRUD) member profiles to effectively manage their club's roster.

**SRS-SCMS-002: Event Management:** The SCMS shall allow Club Administrators to create, manage, and track club events, including setting attendee quotas and specifying event locations.

**SRS-SCMS-003: Financial Tracking:** The SCMS shall provide a financial tracking module that allows Club Administrators to seamlessly record and monitor club income (such as membership dues) and expenditures.

**SRS-SCMS-004: Membership Portal:** The SCMS shall feature a Membership Portal that enables General Members to log in, view their current membership status, and browse a list of upcoming club events.

### 2.2 System Architecture

The system is built on a strict 3-Package Architecture, as defined in the Software Development Plan (SDP). This ensures a clean separation of concerns across the application:

1. **Presentation Layer (UI):** Composed of JavaFX controllers and FXML files. This layer is strictly responsible for capturing user input and displaying data.
2. **Application Layer (Business Logic):** Acts as the "brain" of the application. It handles core SCMS functionality, including member validation, event management rules, and budget calculations.
3. **Data Layer (Database):** Contains Data Access Object (DAO) classes and handles SQL database connections. It interacts solely with the Application Layer, ensuring the UI never directly queries the database.

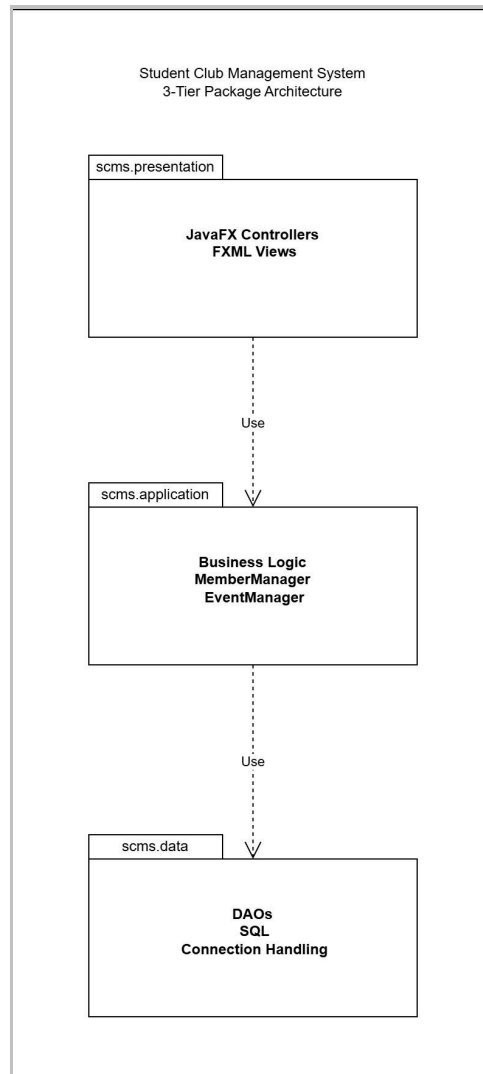


Diagram 1- SCMS Package structure and data flow

## 2.3 Use Cases and Usage Scenarios

### Principal Actors:

- **Administrator (Club President):** Has elevated privileges to manage members, organize events, and track finances.
- **Member (Student):** Has standard privileges to view their personal profile and explore upcoming events.

## Software Requirements Specifications of SCMS software

**Doc # SCMS-SRS**

**Version: 1.5**

**Page 7 / 12**

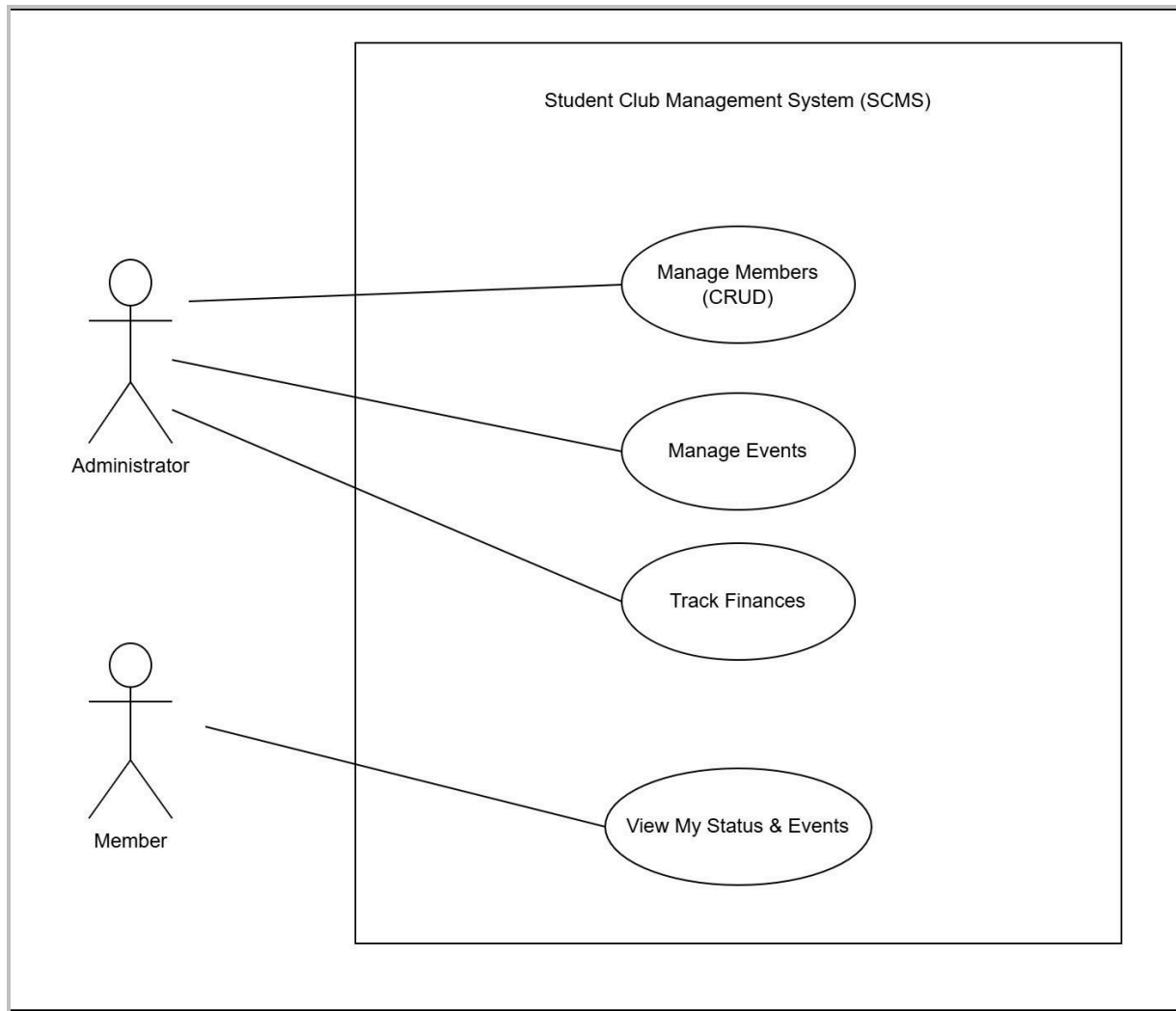


Diagram - 2: SCMS Use Case

| Use Case ID | Use Case Name  | Actor         | Brief Description                                                 |
|-------------|----------------|---------------|-------------------------------------------------------------------|
| UC-1        | Manage Members | Administrator | Create, read, update, and delete member profiles in the database. |

| Software Requirements Specifications of SCMS software |              |             |
|-------------------------------------------------------|--------------|-------------|
| Doc # SCMS-SRS                                        | Version: 1.5 | Page 8 / 12 |

|      |                      |               |                                                                                  |
|------|----------------------|---------------|----------------------------------------------------------------------------------|
| UC-2 | Create Event         | Administrator | Create a new club event with designated capacity limits (quotas) and a location. |
| UC-3 | Record Finance       | Administrator | Log incoming dues and outgoing club expenses to maintain an accurate budget.     |
| UC-4 | View Status & Events | Member        | Access personal membership status and view a list of upcoming events.            |

#### Usage Scenario 1: Administrator creates an event

1. The Administrator navigates to the Event Management dashboard.
2. The Presentation Layer displays the event creation form.
3. The Administrator inputs the event name, date, location, and attendee quota, then submits the form.
4. The Application Layer validates the inputted data to ensure no constraints are violated.
5. Upon successful validation, the Application Layer calls the Data Layer to persist the new event in the database.
6. The system displays a success confirmation and refreshes the upcoming events list.

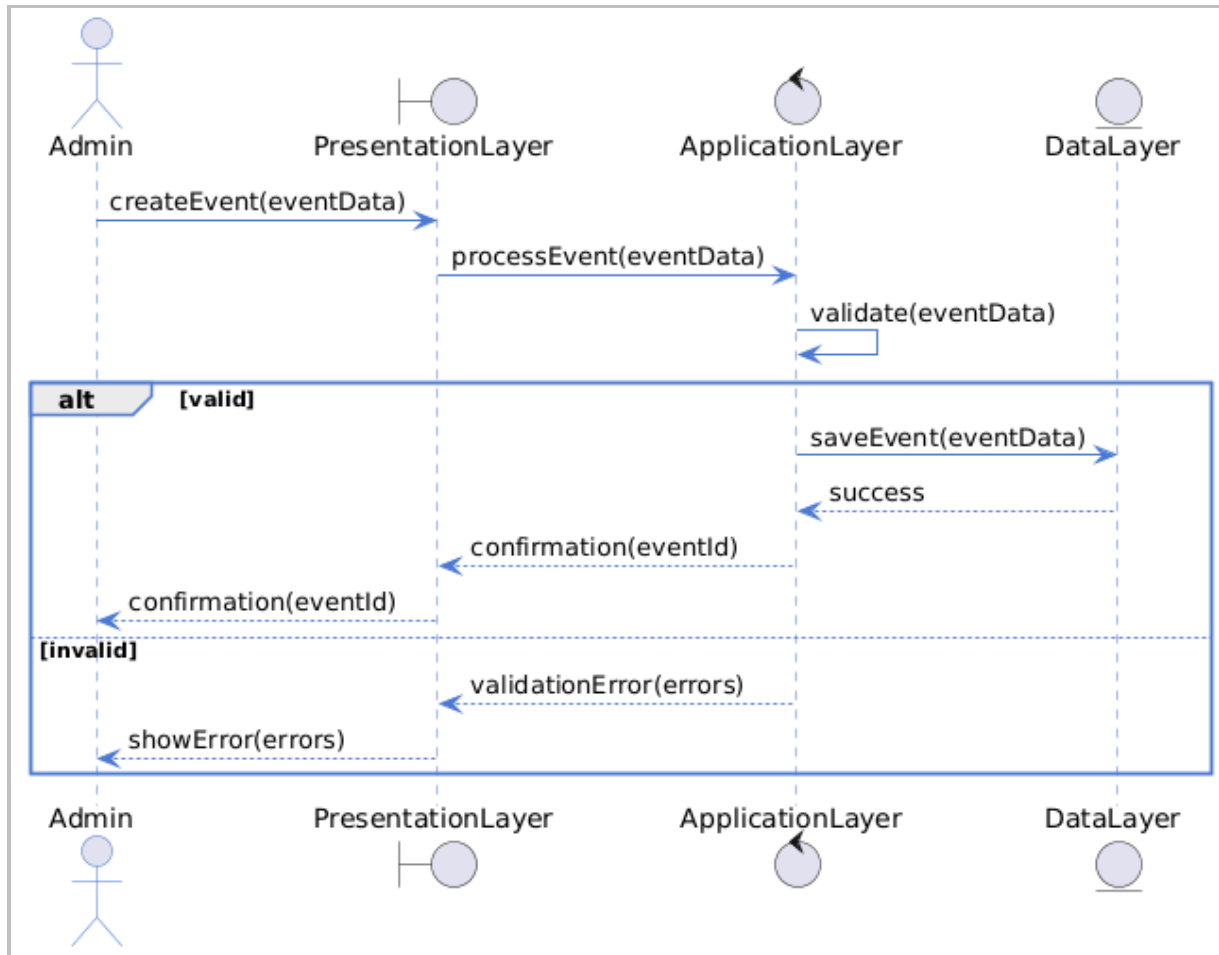


Diagram - 3: Create Event Sequence Diagram

## Usage Scenario 2: Member views upcoming events

1. A Member logs into the SCMS application.
2. The system authenticates the credentials and loads the member's profile.
3. The Member navigates to the "Upcoming Events" tab.
4. The Application Layer requests a list of scheduled events from the Data Layer based on the current date.
5. The system renders the event list on the UI within the required 2-second response time.

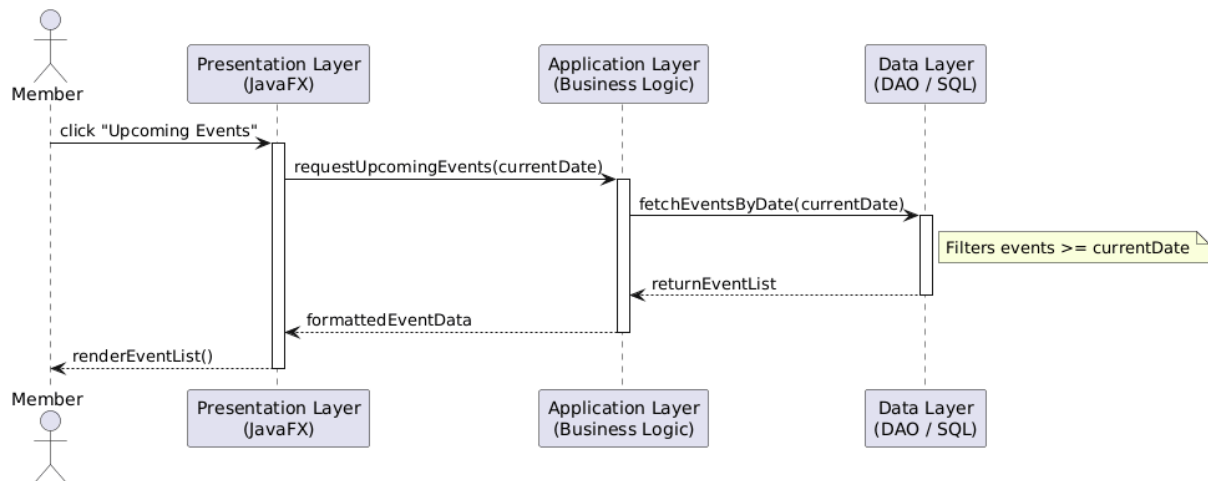


Diagram - 4: Member views upcoming events.

### Usage Scenario 3: Administrator registers a new member

1. The Administrator navigates to the Member Management dashboard.
2. The Presentation Layer displays the member registration form.
3. The Administrator inputs the member's Name, Student ID, Email, and Role, then submits the form.
4. The Application Layer (Member Controller) validates the input format to ensure the Student ID consists of exactly 8 numeric digits (SRS-SCMS-001.2).
5. If the format is valid, the Application Layer calls the Data Layer (Member DAO) to check if the Student ID already exists in the system (SRS-SCMS-001.3).
6. If the ID is unique, the Data Layer persists the new member record, and the system displays a success confirmation. If any validation or uniqueness check fails, an appropriate error message is shown to the Administrator.

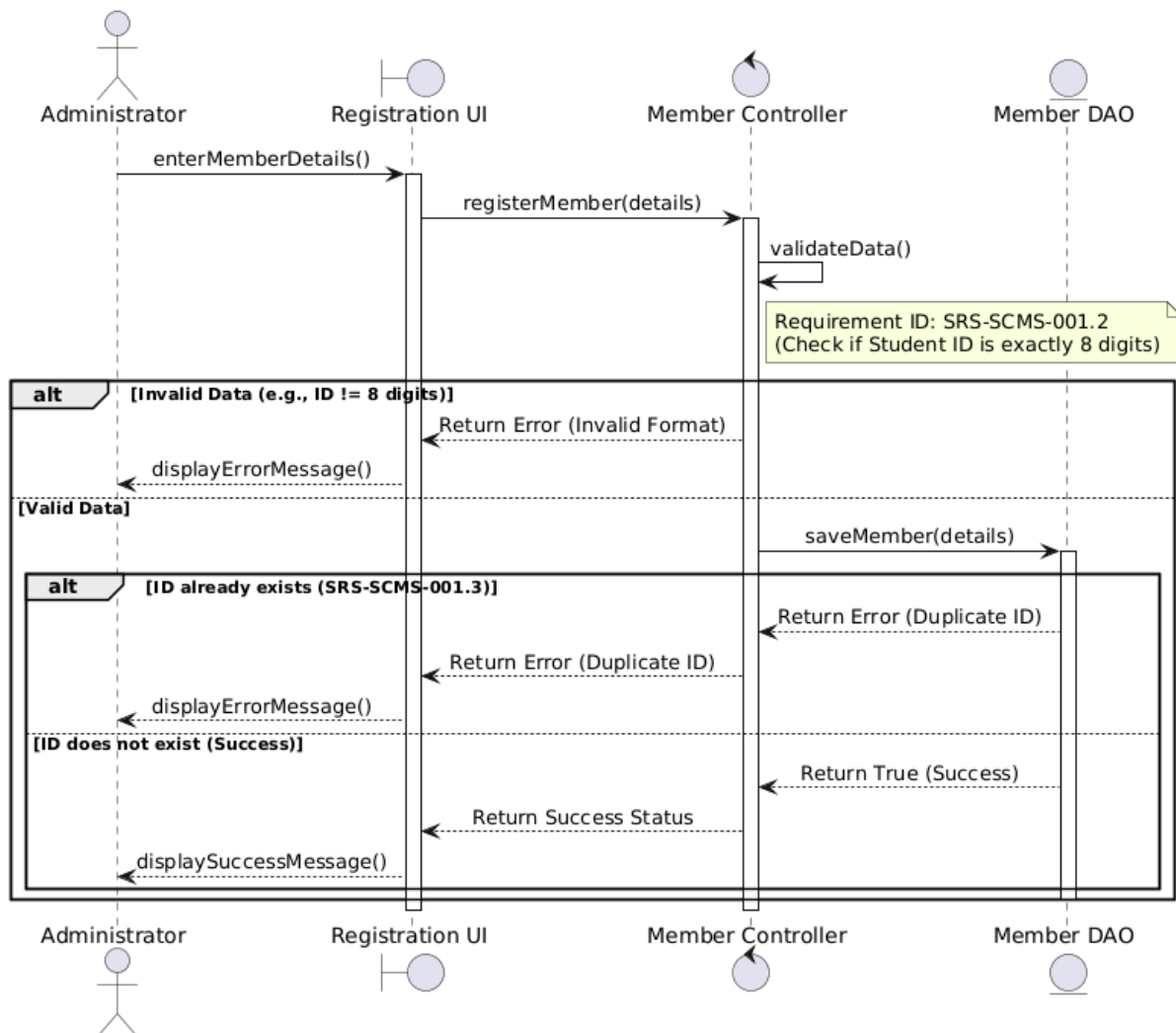


Diagram 5 - Member Registration Sequence Diagram

## 2.4 System Requirements Specification

The following system requirements define the technical, measurable specifications necessary to satisfy the overarching user requirements outlined in Section 2.1.

### Member Management (Parent: SRS-SCMS-001)

- **SRS-SCMS-001.1:** The Presentation Layer shall provide a secure GUI form for Club Administrators to input new member details, specifically requiring: Name, Student ID, Email Address, and Club Role.
- **SRS-SCMS-001.2:** The Application Layer shall validate that the submitted Student ID consists of exactly 8 numeric digits before permitting the creation of a new member record.
- **SRS-SCMS-001.3:** The Data Layer shall enforce a unique constraint on the Student ID field to prevent duplicate member registrations within the same club.

| Software Requirements Specifications of SCMS software |              |              |
|-------------------------------------------------------|--------------|--------------|
| Doc # SCMS-SRS                                        | Version: 1.5 | Page 12 / 12 |

#### Event Management (Parent: SRS-SCMS-002)

- **SRS-SCMS-002.1:** The system shall allow Administrators to define an event with the following required parameters: Event Name, Date, Location, and Maximum Attendee Quota.
- **SRS-SCMS-002.2:** The Application Layer shall track the number of member sign-ups for an event and automatically reject any registrations that exceed the defined Maximum Attendee Quota.
- **SRS-SCMS-002.3:** The system shall present an error message to the Administrator via the GUI if they attempt to schedule an event in the past.

#### Financial Tracking (Parent: SRS-SCMS-003)

- **SRS-SCMS-003.1:** The Application Layer shall automatically calculate and update the club's available remaining budget immediately after any income or expense entry is recorded.
- **SRS-SCMS-003.2:** The Data Layer shall persistently store all financial transactions, logging the exact amount, date of transaction, and a text description for auditing purposes.
- **SRS-SCMS-003.3:** The Presentation Layer shall feature a real-time dashboard for Administrators displaying the aggregated total income, total expenses, and the calculated remaining balance.

#### Membership Portal (Parent: SRS-SCMS-004)

- **SRS-SCMS-004.1:** The system shall retrieve and display the list of "Upcoming Events" to the Member within 2 seconds of a successful login authentication.
- **SRS-SCMS-004.2:** The Presentation Layer shall restrict Members to viewing only their personal profile data and the public event calendar, hiding all administrative controls (CRUD operations on other members, budgeting).
- **SRS-SCMS-004.3:** The Data Layer shall filter the event queries so that Members only see events scheduled for the current date or future dates.

#### Non-Functional Requirements

- **SRS-SCMS-NF-01:** The Presentation Layer shall be developed utilizing the JavaFX framework to ensure a responsive and modern graphical user interface.
- **SRS-SCMS-NF-02:** The codebase shall strictly adhere to Oracle Java Code Conventions and maintain the separation of concerns dictated by the 3-Package Architecture (Presentation, Application, Data).
- **SRS-SCMS-NF-03:** All Data Layer interactions with the SQL database shall strictly utilize parameterized queries (Prepared Statements) to mitigate the risk of SQL injection attacks.